

The Zhuanyu Calendar and the Archaeological Evidence for the Calendars of Qin and the Early Han

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Abstract

This essay is about a calendar known almost entirely from a single sentence. The histories record that Qin Shi Huang, having unified the realm, adopted the 「颛顼历」, the Zhuanyu calendar, and fixed the head of the year at the tenth month; and that the Han took the same calendar over without change until the Taichu reform of 104 BCE. What the calendar itself looked like — how many days the year held, which months were long and which short, where the intercalary month was placed, which day began a month — the transmitted texts do not say. Excavated bamboo slips do. This piece sets the three textual statements against three groups of excavated calendar manuscripts: the calendar for the thirty-fourth year of Qin Shi Huang (213 BCE) from tomb no. 30 at Zhoujia tai in Jingzhou; the eighteen-slip calendar from tomb no. 247 at Zhangjiashan, running from the fifth year of Han Gaozu to the second year of Empress Lü (202–186 BCE); and the thirty-two-slip table from tomb no. 2 at Yinqueshan, for the first year of Yuanguang (134 BCE), which corrects an error carried for a thousand years by the Song Zizhi tongjian mulu. It explains why the intercalary month was called 「后九月」, the "latter ninth month"; sets out the Qin–Chu month-name concordance from the Shuihudi Rishu; and separates what the slips can be shown to support from what remains inference. Where a point is one view among others rather than a general one, it is labelled 【一说】 (one view). The piece is descriptive and carries no prediction.

Keywords: Zhuanyu calendar; 颛顼历; Qin calendar; early Han calendar; 后九月; intercalary month; bamboo slips; Zhoujia tai; Zhangjiashan; Yinqueshan; Shuihudi; Qin and Chu month names; Chinese calendrics

Introduction

At the new year, the men of Qin celebrated in the tenth month.

This is written in the histories. When Qin Shi Huang unified the realm he used the 「颛顼历」, the Zhuanxu calendar; the head of the year was fixed at the tenth month, and the court's congratulations were reckoned from the first day of that month. The Han took over and did not change it, and the calendar stayed in use down to the first year of Emperor Wu's Taichu reign.

That one sentence is all the transmitted texts yield. What the calendar itself looked like cannot be seen. How many days the year held, which months were long and which short, where the intercalary month was placed, which day began a month — none of it.

Then, batch after batch, bamboo slips came out of tombs. One had the sexagenary day-signs for the whole of the thirty-fourth year of Qin Shi Huang copied out; one set the first days of the months, day by day, from the fifth year of Han Gaozu to the second year of Empress Lü; and one was made into a table, laid out on slips sixty-nine centimetres long.

Only now does the sentence on paper acquire something underneath it.

1. A name known only from paper

Take the name first.

The 「颛顼历」 is one of the 「古六历」, the six ancient calendars. These are the calendars attributed to Huangdi, Zhuanxu, the Xia, the Shang, the Zhou and the state of Lu, listed side by side in the *Hanshu*, 「Lǚli zhi」. Of the six, the one actually taken up as the official calendar of a unified dynasty is the Zhuanxu.

How it was laid out, however, has always rested on indirect testimony. The old writers say it was a 「四分历」, a quarter-day calendar, with a year of three hundred and sixty-five and a quarter days. That number was worked out in the previous piece, on the gnomon: measure the winter-solstice shadow over four consecutive years, and the shadow comes back to its starting point after one thousand four hundred and sixty-one days; divide by four and there it is. The name 「四分历」 comes from that fraction.

The *Huainanzi*, 「Tianwen xun」, of the early Han adds a line:

反覆三百六十五度四分度之一而成一岁……天一以始建七十六岁，日月复以正月入营室五度无余分，名曰一纪。

Seventy-six years, and sun and moon come back to their places; this is called one 「纪」, one epoch. The Zhuanxu calendar rests on that single number.

But a number that can be derived on paper is not the same as the number the calendar officers of the day actually used. To establish that, one has to look at what they wrote down themselves.

2. The evidence on paper

Three statements, first, out of the transmitted texts.

The first is the *Shiji*, 「Qin Shihuang benji」:

始皇推终始五德之传，以为周得火德，秦代周德，从所不胜。方今水德之始，改年始，朝贺皆自十月朔。

「改年始」 means the head of the year was changed; 「朝贺皆自十月朔」 means the court's congratulations were reckoned from the first day of the tenth month. Zhang Shoujie, commenting in the Tang, glosses the line:

周以建子之月为正，秦以建亥之月为正，故其年始用十月而朝贺。

The Zhou took the month of 建子 as the first month, the Qin took the month of 建亥 as the first month, and so the head of the year fell in the tenth month.

The second is the *Shiji*, 「Zhang Chengxiang liezhuan」. Zhang Cang, chancellor in the early Han, had served under the Qin as an imperial clerk, in charge of the palace books and of pitch-pipes and the calendar. When the Han was founded he put the new dynasty's calendar in order:

张苍为计相时，绪正律历。以高祖十月始至霸上，因故秦时本以十月为岁首，弗革。

Liu Bang entered the passes in the tenth month and reached Bashang; Zhang Cang held that since the old Qin had taken the tenth month as the head of the year, it should not be altered. At the end of the same chapter Sima Qian appends a remark:

张苍文学律历，为汉名相，而绌贾生、公孙臣等言正朔服色事而不遵，明用秦之颛顼历，何哉？

Jia Yi and Gongsun Chen and others urged a reform of the calendar and of the first day of the month; Zhang Cang would hear none of it, and 「明用秦之颛顼历」, plainly used the Zhuanxu calendar of the Qin. This rhetorical question is one of the few appearances the Zhuanxu calendar makes in the *Shiji*.

The third is the *Hanshu*, 「Lüli zhi」. Ban Gu states the matter most fully:

汉兴，方纲纪大基，庶事草创，袭秦正朔。以北平侯张苍言，用《颛顼历》，比于六历，疏阔中最为微近。然正朔服色，未睹其真，而朔晦月见，弦望盈亏，多非是。

「袭秦正朔」: the Qin's reckoning was inherited. 「用《颛顼历》」: that is the calendar used. 「比于六历, 疏阔中最为微近」: against the other five it is the crudest, and yet the closest of them to the actual sky. The last clause matters: used long enough, its first days of the month no longer fell on the new moon, the moon was still visible on the last day of a month, and the dates of the quarters were often wrong. This is the ground on which Emperor Wu afterwards reformed the calendar.

Three statements on paper, all pointing at one thing: Qin and the early Han used a calendar called the Zhuanxu, with the tenth month at the head of the year. How it was laid out, the texts still do not say.

3. The evidence from the ground: Qin

In the first half of 1993, at Guanjü township in the Shashi district of Jingzhou, Hubei, an archaeological team excavated a Qin tomb, numbered no. 30 at Zhoujiatai. Three hundred and eighty-seven bamboo slips were recovered from it.

Among them is a calendar manuscript, a copy of the sexagenary day-signs for the whole of the thirty-fourth year of Qin Shi Huang, that is, 213 BCE.

Its layout is a different matter from a modern wall calendar. The slips are a little over twenty-nine centimetres long, and a single slip is divided lengthwise into six sections, one section to a month. Six short months are copied first — the tenth, twelfth, second, fourth, sixth and eighth months, twenty-nine days each; then six long months — the eleventh, first, third, fifth, seventh and ninth, thirty days each. Short and long alternate the whole way down. At the end one further month is added, the 「后九月」, thirty days.

Twelve months plus one intercalary month come to three hundred and eighty-four days. In that year the extra month hung after the ninth.

The slips carry other characters besides. Alongside the day-signs there are scattered notes on the tomb occupant's movements: on such a day he lodged at Jingling, on such a day at Jiangling, on such a day at Huangyou, on such a day he conducted business at Jingling; and in the first month there is an entry 「嘉平视事」. Researchers read these as the man's journeys and his office work. A calendar manuscript, used as a notebook.

From the same tomb came a wooden tablet, with the first days of the months of the first year of Qin Er Shi (209 BCE) on its face, the day-signs of the twelfth month on its back, and characters of the sort 「廷赋所」 at the top. In the same batch of slips are the month-beginnings and month-lengths for the thirty-sixth and thirty-seventh years of Qin Shi Huang; the one for the thirty-sixth year carries a title after it, reading 「卅六年日」. People of that time gave their calendar manuscripts names.

4. The evidence from the ground: the early Han

At the end of 1983, at Zhangjiashan in Jiangling, Hubei, tomb no. 247 of the Han yielded a batch of bamboo slips. Eighteen of them, each twenty-three centimetres long, carried no original title; the editors gave them one and called them the 「历谱」, the Calendar Tables.

Eighteen slips, one to a year. Each, from top to bottom, gives the first days of the twelve months and of the intercalary month of that year. The first to the ninth are the fifth to the twelfth years of Han Gaozu (202 BCE onward); the tenth to the sixteenth are the first to the seventh years of Emperor Hui; the seventeenth and eighteenth are the first and second years of Empress Lü. The latest entry reaches the 「后九月」 of Empress Lü's second year.

This is the earliest practical calendar manuscript of the early Western Han yet found. "Practical" is a word with weight: it is not a table computed by later men, but something copied down and used by people of the time.

Further on, in April 1972, tomb no. 2 of the Han at Yinqueshan in Linyi, Shandong. The calendar manuscript from this tomb is more carefully made than the two before it.

There are thirty-two slips, sixty-nine centimetres long, which on the Han foot is exactly three chi. The first slip records the year, the second the months, and from the third onward one day to a slip. Laid out, the thirty-two slips are a table.

It records thirteen months. From the tenth month, then the eleventh, the twelfth, the first ... all the way to the ninth, with one more added at the end, the 「后九月」.

Find-spot	Date	Form	Content
Tomb no. 30, Zhoujiaitai, Jingzhou, Hubei	34th year of Qin Shi Huang (213 BCE)	bamboo slips, one divided into six sections	sexagenary day-signs for the year, short and long months alternating, ending with a 后九 月
Tomb no. 247, Zhangjiashan, Jiangling, Hubei	Han Gaozu 5 (202 BCE) to Empress Lü 2 (186 BCE)	eighteen bamboo slips, 23 cm long	one slip per year, the first days of the months given from top to bottom
Tomb no. 2, Yinqueshan, Linyi, Shandong	1st year of Yuanguang, Emperor Wu (134 BCE)	thirty-two bamboo slips, 69 cm long	a table read across, thirteen months, with calendar notes

The manuscript also carries calendar notes. Besides the days, the slips mark the 腊日, the winter solstice, the summer solstice, the beginning of spring and the beginning of autumn, along with the first, middle and last 伏, and a kind of day called 「反支」.

The most important thing about it is that it corrected an error. This manuscript records that the twelfth month and the first month of that year were both thirty days long, two long months running; the Song Zizhi tongjian mulu recorded them as short months. For more than a thousand years readers of history followed that book, and only after the slips came out was the reading put right.

The first slip carried a title whose characters are read in two ways; some read 「七年视日」, and the editors proposed a name of their own, the 「元光元年历谱」, the Calendar Tables for the First Year of Yuanguang. Emperor Wu's 「建元」 reign had no seventh year: the seventh year of Jianyuan had in fact already changed the era, and was the first year of Yuanguang (134 BCE).

5. Where the 「后九月」 comes from

All three of the manuscripts above contain a 「后九月」, a "latter ninth month". This month-name needs a word of its own.

The moon goes round in a little over twenty-nine and a half days, and twelve months come to about three hundred and fifty-four days; the sun goes round in a little over three hundred and sixty-five. The ten-odd days of difference, gathered over two or three years, make up a whole month. That month has to be inserted, and it is called the intercalary month.

Where to insert it differed from house to house. Under the Qin and in the early Han the intercalary month was put at the end of the year and called 「后九月」. The ancients had a name for the practice: 「归余于终」, returning the surplus to the end. The intercalation cycle of the quarter-day calendar was seven intercalary months in nineteen years.

Why should the end of the year be the ninth month? Because the head of the year was the tenth. The year began in the tenth month, so its end naturally fell in the ninth, and the extra month hung after the ninth; such a year had thirteen months.

The term 「后九月」 appears very early. Among the Shuihudi Qin slips there is a text called the 「编年纪」, which records major events year by year. Under the fifty-sixth year of King Zhao of Qin it has a single line:

五十六年，后九月，昭死。

King Zhao of Qin died in that year, and the month is given as the 「后九月」. The fifty-sixth year of King Zhao is 251 BCE. The month-name was already in use by then.

The Japanese scholar Kurihara Tomonobu saw the same thing in the order of entries in the Shiji, 「Qin benji」: under the forty-second year of King Zhao the tenth month is recorded before the ninth, and under the forty-eighth year the tenth month before the first. 【一说】

From this he inferred that taking the tenth month as the head of the year did not begin with the twenty-sixth year of Qin Shi Huang. Since the slips came out, that point has had physical objects alongside it.

6. Two sets of month names, Qin and Chu

What has been said so far concerns the months of Qin. But the lands the men of Qin conquered had belonged to other states. In Chu the months were called by another set of names.

In December 1975, a batch of bamboo slips came from tomb no. 11 of the Qin at Shuihudi in Yunmeng, Hubei. Among them, in the 「日书」, the *Rishu*, in its first recension, is a concordance of Qin and Chu month-names, month by month:

Qin month	Chu month	Qin month	Chu month
tenth	冬夕	fourth	七月
eleventh	屈夕	fifth	八月
twelfth	援夕	sixth	九月
first	刑夷	seventh	十月
second	夏	eighth	爨月
third	纺月	ninth	献马

The tenth month of Qin is called 冬夕 in Chu; the eleventh, 屈夕; the twelfth, 援夕. The names further down look even less like month-names: 刑夷, 纺月, 爨月, 献马.

These Chu month-names were not known to be month-names at all until recently. The inscription on the bells from Changtaiguan in Xinyang contains characters of the form 「屈栾」, which had never made sense; once the concordance appeared, it became clear that they were the month-names of Chu.

The table says one thing: Qin took the tenth month as the head of the year, but the months themselves still followed the Xia reckoning. The first month was still the first month; whichever month the handle of the Dipper pointed to at 建寅 was still called the first month. The difference between the months of Qin and of Chu lay exactly there.

7. Whose business the calendar is

The preceding sections have dealt with what can be seen on the slips. There is one more layer, written outside them.

Why did Qin take the tenth month as the head of the year? The *Shiji*, 「Qin Shihuang benji」, puts the reason in the virtue of the dynasty, not in the sky. The passage is about the virtue of

Water: the Zhou had the virtue of Fire; the Qin replaced the Zhou, and so took the one that overcomes Fire, which is Water. At the beginning of the virtue of Water, therefore, 「改年始」, the head of the year was changed; therefore the court's congratulations ran 「皆自十月朔」; therefore clothes and banners were 「皆上黑」; therefore 「数以六为纪」, the number six was made the measure.

Where the first day of the year is moved to is a posture; it tells the realm that the dynasty has changed from the ground up.

That the early Han did not change it was also a posture. Zhang Cang, putting the new dynasty's calendar in order, kept the old Qin calendar on one hand and worked out the five virtues on the other, holding that the Han too had the virtue of Water, and therefore 「尚黑如故」, still honoured black as before. He took the whole Qin apparatus over as it stood. 【一说】 The doctrine of the Water virtue was later challenged in a special study, which argued that "the Qin used the virtue of Water" was an elaboration by later men. It is recorded here for what it is.

Politics aside, the calendrical facts can still be counted. The Zhuanxu calendar was in use from the year of Qin's unification (221 BCE) to the reform of the calendar in the seventh year of Yuanfeng under Emperor Wu (104 BCE), a span of one hundred and seventeen years. From the founding of the Han to the reform, Ban Gu records 「汉兴百二岁矣」, a hundred and two years.

More than a century on one calendar, until the first days of the months no longer came out right and the full and waning moons were wrong, before the change was allowed.

8. What can be checked, and what is only inference

Looking back, the texts above ground and the objects below it together establish this much.

Qin and the early Han used a calendar called the Zhuanxu, with the tenth month at the head of the year. Three statements on paper and three groups of slips agree. The months of the year ran from the tenth to the ninth; the intercalary month was placed after the ninth and called the 「后九月」; the months alternated long and short, with one 后九月 added at the end. All this is visible on the Zhoujiaitai and Yinqueshan manuscripts alike. The month-names of Qin and those of Chu are two different sets.

What cannot be checked should also be set out.

The slips are broken. The Zhoujiaitai manuscript copies the thirty-fourth year; the same tomb holds month-beginnings for the thirty-sixth and thirty-seventh, so the years join up, but the gaps between them cannot be filled. The eighteen Zhangjiashan slips stop at the 「后九月」 of Empress Lü's second year, and the stretch after that has to be picked up from other materials. Calendars in use in different places need not have been the same; within one batch of Qin slips

there is more than one case of day-signs differing by a day, and some researchers hold that this is not necessarily a mistake but may reflect the priority of the calendars relied on.

What algorithm actually sits under the name "the Zhuanxu calendar" is likewise inference. The quarter-day year, the seven intercalations in nineteen years, the seventy-six-year epoch — these numbers can be read out of later literature; whether the copy in the hands of the calendar officers of the day was accurate, the slips say not a word.

The man in the tomb, copying out his calendar manuscript, most likely never thought of leaving it for people two thousand years later. He was writing down which day he travelled and which day he worked.

Note on the text

This is a popular exposition, and it makes no claim to original research. The calendrical facts cited rest on transmitted texts and on excavated material. This English version is issued alongside the Chinese original as the same article in another language. The author's name is 卞楚嘉, Chujia Kuang; he writes under the pen-name 嘉言一得 (Jiayan Yide).

Chinese terms are left in their original characters where the word itself is the subject of the discussion.

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